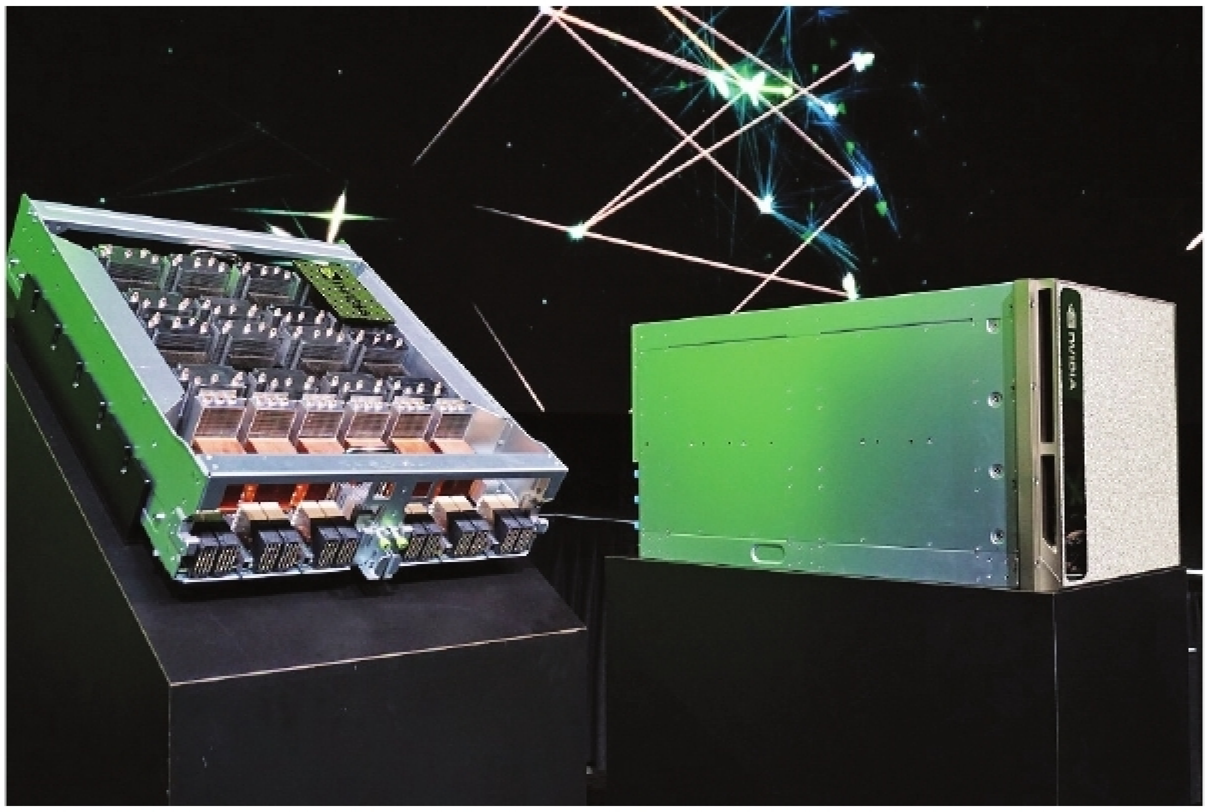
Self-driving cars is one of the biggest applications of AI, covering aspects of safety and easing traffic congestion. NVIDIA is being generous with its technology in this field. The company is not only working on its own self-driving technology, but is also working with other companies like Uber, Baidu, Toyota, Mercedes-Benz, Audi, Volkswagen and Tesla. Clearly, it believes that partnering is the way forward.

A big thrust area for this company, moving forward, is the supercomputers that are driving all the cloud services. Amazon, Microsoft, Google, Baidu and others are all giving up using CPU servers and are moving to GPU servers. Just to give you an idea of how much more powerful a GPU server is compared to a CPU server—one GPU server can replace 160 CPU servers. And when you interconnect multiple GPUs, it can work at a speed of 2 petaFLOPS (for the nerds, that is 1015 FLOPS). Such computers are needed to solve the most complex AI challenges of the future and frankly, it is mindboggling to think about where GPUs are taking the power of computing.

Takeaways: Understanding the market dynamics as well as its consumers was critical to NVIDIA's success. Its strategy to continuously provide product upgrades at six-month intervals and to keep innovation as its core function has propelled it to great heights. The company has pushed its product designs to the maximum technical limits, always providing the best end user performance. NVIDIA also chose its niche well—gamers. This audience was smaller than the main consumer market of PC users, but members from this group were knowledgeable and that has kept NVIDIA at the cutting-edge of technology. NVIDIA has found the right forums to connect with them either at conferences or through social media, and used tech influencers well. But perhaps the best is yet to come. The age of supercomputers and supercomputer processors has put AI on the fast track and NVIDIA is at the centre of this revolution. Will it be the next Intel?

Zeiss

Zeiss has been the camera of choice for generations and was the ingredient brand for Nokia phones in the 2000s as well. Carl Zeiss opened his workshop in 1846, focusing on making scientific instruments and lenses. He developed the famous Jena optical glass. By 1861, the company was considered one of the best in Germany. By WW1, it was the world's largest camera production company. And in 1969, the Zeiss Biogon lens was used in the Apollo 11 mission to the moon by NASA. That is the heritage of Zeiss —always considered a pioneer and leader in lenses.



NVIDIA DGX-2 with the computing power to deliver two petaFLOPS

ZEISS

It's not that Zeiss had no competition over the years. It's just that it was more focused on addressing consumer needs and on keeping up with the market. Over the years, Zeiss also built a brand. It developed and released campaigns showcasing its new lenses and the picture quality from Zeiss lenses.

Zeiss Biogon lenses

Zeiss also entered into collaborations and licensing agreements with many other companies like Hasselblad, Rollei, Yashica, Sony, Logitech and Alpha. It either entered into co-branding agreements for lenses that were developed by other companies, or it licensed its technology to other companies for complete optical design and manufacturing.

Zeiss and Nokia: By 2004, Nokia phones were the highest selling in the world. Yet, Nokia did not have the best phone camera. Sony Ericsson did. And so Nokia roped in Zeiss to develop a lens for its mobile phone. The result was the N90 launched in April 2005 with a 2MP camera by Zeiss.

Under this partnership, the two companies produced a series of phones, including the Nokia N95 with a 5MP camera, and had a steady hold over the consumer market. A milestone of the Nokia and Carl Zeiss partnership was the launch of the Nokia 808 PureView phone in 2012. The highlight was the name of the phone, which was solely based on its imaging quality.